

Payroll Tax Relief and Labour Market Informality: Evidence from Brazil

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Abstract

This paper estimates the effect of Brazil’s 2011 payroll tax relief program – the *desoneração da folha* – on labour market informality at the sector level. Using microdata from the Pesquisa Nacional por Amostra de Domicílios Contínua (PNADc) and exploiting the staggered rollout of the reform across sectors, I apply the imputation estimator of Borusyak et al. (2024) to a panel of sector-quarter observations spanning 2012 to 2014, benchmarked against three alternative estimators. The estimates point to a positive but statistically insignificant average treatment effect on the informality share, consistent across all four specifications considered. This null result aligns with existing evidence suggesting that informality is not a primary margin of adjustment to payroll tax changes in Brazil. Beneath the aggregate null, cohort-specific estimates reveal meaningful heterogeneity: the two treated cohorts point in opposite directions, with the overall null reflecting their weighted average rather than a uniform zero effect. The dynamic pattern warrants further investigation, but its causal interpretation remains uncertain given the data limitations inherent to household survey data.

1 Introduction

Brazil’s labour market is characterised by high and persistent informality: roughly a third of workers operate outside the formal sector, without labour protections, social security contributions, or access to the benefits these confer. Reducing informality has long been a stated objective of Brazilian labour and tax policy, and the link between labour costs and the formal-informal margin has motivated successive reforms aimed at lowering the cost of formal employment. In December 2011, the federal government enacted the *desoneração da folha*, a payroll tax relief programme that replaced the standard 20% payroll tax on the wage bill with a substantially lower tax on gross revenue for firms in eligible sectors. The reform was rolled out in a staggered fashion across 2011-2014, ultimately covering a broad set of manufacturing and service sectors. While the policy’s official motivation centred on international competitiveness and formal job creation, its implications for the broader labour market, and in particular for informality, remain an open empirical question.

A large international literature has examined the labour-market effects of payroll taxation. [Gruber \(1997\)](#) finds near-complete pass-through of payroll taxes to wages in Chile, with limited employment effects, while [Saez et al. \(2019\)](#) document significant employment responses to a payroll tax cut targeted at young workers in Sweden, concentrated among small firms. A smaller body of work in developing countries has examined the informality margin specifically. [Kugler and Kugler \(2009\)](#) find that payroll tax increases in Colombia reduced formal employment and shifted workers toward informality. [Kugler et al. \(2017\)](#) document that a subsequent Colombian payroll tax cut produced substantial formalisation effects, with the probability of having a written employment contract increasing by 8.5 percentage points among eligible workers. Together these studies establish that the incidence of payroll taxes depends critically on the competitive structure of the labour market and the design of the policy, with effects varying substantially across institutional settings. This contrasts with the broader structural literature surveyed in [Ulyssea \(2020\)](#), which concludes that reducing the costs of formality generates limited responses on the informality margin, even when ag-

gregate effects on output and productivity are positive. The evidence on whether payroll tax cuts meaningfully shift labour market composition therefore remains mixed, with effects depending on the institutional setting and the relevant time horizon.

The theoretical case for informality responses to payroll taxation rests on the fact that informality is multidimensional. [Ulyssea \(2018\)](#) distinguishes the extensive margin of unregistered firms, the intensive margin of unregistered workers within registered firms, and self-employment as distinct responses to the costs of formality, with fixed costs of formalisation generating heterogeneous firm-level adjustments. [Haanwinckel and Soares \(2021\)](#) develop a complementary search-and-matching framework in which firm and worker compliance decisions, together with labour regulations and the minimum wage, jointly determine the equilibrium informality rate. Both frameworks imply that payroll tax cuts can affect informality, with the magnitude depending on the structure of fixed costs, the extent of wage adjustment, and the level of within-sector heterogeneity.

The Brazilian literature on the *desoneração da folha* has approached the policy from several complementary angles. [Baumgartner et al. \(2022\)](#) use linked employer-employee administrative data and a staggered difference-in-differences design to estimate a 5% increase in formal employment in treated sectors, with no detectable wage effects, and treat sectoral informality as a heterogeneity dimension rather than as an outcome. [Lobel \(2024\)](#) provides the most complete analysis of the reform’s incidence, using linked administrative records to estimate employment, wage, and profit effects and to decompose the burden across consumers, workers, and firm owners. In examining the origins of new formal hires, he finds that the share coming from informality or non-employment does not change with treatment, and concludes that formalisation is not a significant margin of response. [Meyer et al. \(2021\)](#) compare quasi-experimental and structural estimates of the policy’s effect on formal employment and conclude that reduced-form approaches likely overstate its impact relative to structural counterfactuals. Common to all three contributions is reliance on administrative data, which by construction observes only the formal sector and cannot directly measure

aggregate labour market composition outside the formal sector.

This paper contributes to the literature on the payroll tax cut by providing sector-level evidence on the informality margin using household survey data from the Pesquisa Nacional por Amostra de Domicílios Contínua (PNADc). PNADc observes labour market status regardless of formality, allowing direct measurement of the informality share among employed workers¹, specifically the share of workers without formal contracts among private sector employees – the dimension of informality that corresponds most closely to the intensive margin in the sense of [Ulyssea \(2018\)](#), whereby formally registered employers hire workers off the books. Administrative data, which records only formally contracted employment relationships, cannot directly measure this margin. The treatment definition follows [Baumgartner et al. \(2022\)](#), restricting attention to sectors made eligible under the ISIC criterion and applying their treatment assignment to a panel of sector-quarter observations spanning 2012 to 2014. The contribution is complementary to existing work rather than competing with it. [Lobel \(2024\)](#) examines the worker-level origins of new formal hires using administrative records, an analysis conditional on appearing in formal employment during the sample period. Building on this evidence, this paper looks at the aggregate informality share at the sector level using survey data that observes the working-age population irrespective of formal-sector participation. This focus on a different object – the share of informal employment among the employed, rather than the inflow composition of new formal hires – provides a complementary view of the policy’s effect on labour market composition along the informality margin.

The empirical strategy applies the imputation estimator of [Borusyak et al. \(2024\)](#) as the preferred specification, benchmarked against [Callaway and Sant’Anna \(2021\)](#), [de Chaisemartin and D’Haultfoeuille \(2024\)](#), and the canonical two-way fixed effects estimator. Overall ATT estimates are positive but statistically insignificant across all specifications, consistent with existing evidence suggesting that informality is not a primary margin of adjustment

¹Indeed, the Brazilian Institute of Geography and Statistics (IBGE) itself uses PNADc microdata to compute and publish Brazil’s official labour informality rate.

to payroll tax changes in Brazil. Beneath this aggregate null, however, cohort-specific estimates reveal meaningful heterogeneity: cohort 7 ATTs are positive and significant under some specifications while cohort 3 ATTs are negative, with the overall null reflecting their weighted average rather than a uniform zero effect. Given the data limitations inherent to household survey data, the estimates capture the evolution of the informality share among the employed rather than speaking directly to the underlying mechanism.

The remainder of the paper is organised as follows. Section 2 describes the institutional setting of the payroll tax relief. Section 3 introduces the PNADc data, sample construction, and outcome definitions. Section 4 sets out the empirical strategy and estimator choices. Section 5 reports the main results, heterogeneity, and a comparison across alternative estimators. Section 6 concludes.

2 Institutional Background

Brazil’s payroll tax system is designed to fund social security programs, including retirement pensions and unemployment insurance, and follows a structure broadly similar to other OECD countries. The main component is a flat 20% tax levied on firms’ total wage bill, with additional components for accident risk insurance and an employee-specific contribution. These taxes are collected exclusively from firms and deposited into a pooled social security fund, meaning workers do not hold individual accounts traceable to their contributions.

In December 2011, the Brazilian government enacted the *desoneração da folha*, a payroll tax relief program introduced under the broader Plano Brasil Maior (PBM) industrial policy framework. The reform replaced the standard 20% payroll tax on the wage bill with a tax of 1% to 2.5% on gross revenue for eligible firms. The official motivation was to enhance the international competitiveness of Brazilian firms, particularly in labour-intensive sectors exposed to foreign competition. The reform was introduced via an executive bill, bypassing prior congressional deliberation, and implementation began immediately for an initial set of

sectors upon enactment.

Eligibility for the tax relief was determined through two distinct criteria. Under the ISIC criterion, all firms in a designated 4-digit sector were automatically eligible. Under the NCM criterion, firms producing specific products listed under the Mercosur Common Nomenclature were partially eligible, with the tax waiver applied proportionally to the share of eligible products in gross revenue. This paper restricts attention to sectors eligible exclusively under the ISIC criterion, for two reasons. First, in sectors with NCM inclusions, treatment status varies across firms within the same sector based on their product mix, a characteristic that is endogenous to firm-level decisions and correlated with productivity and labour demand. This within-sector variation in treatment precludes clean identification of the policy effect. Second, since PNADc does not allow observation of firm-level treatment status, the NCM partial eligibility criterion cannot be implemented at the worker level regardless. Restricting to ISIC-only sectors ensures that all firms in an eligible sector face the same tax change, making sector-level treatment assignment both credible and operationally feasible.

The reform was introduced through a series of legislative acts from 2011 to 2014, progressively extending eligibility to new sectors. The sample used in this paper captures two treated cohorts defined by ISIC-criterion eligibility: an earlier cohort first treated in September 2012 and a later cohort first treated in July 2013. Sectors never granted ISIC eligibility serve as the control group. Treatment is absorbing within the sample period, meaning once a sector gained eligibility it retained it through the end of 2014. From 2015 onwards, the government began a series of legislative changes that progressively altered the taxation structure and ultimately made participation in the program optional for eligible firms. This creates two concerns for identification: first, the treatment status of sectors becomes ambiguous as firms can opt out; second, since opt-out decisions are likely correlated with firm-level characteristics such as profitability and labour demand, treatment assignment from 2015 onwards would no longer be exogenous at the firm level. Even if sector-level eligibility remains defined, the effective treatment intensity within sectors becomes endogenous, making credible

identification infeasible with PNADc data. The analysis is therefore restricted to the twelve quarters from Q1 2012 to Q4 2014.

3 Data

This paper uses microdata from the PNADc, Brazil’s continuous household labour force survey administered by the Brazilian Institute of Geography and Statistics (IBGE). PNADc follows a rotating panel design in which one fifth of the sample is replaced each quarter, with each individual observed for up to five consecutive quarters. The sample covers twelve quarters from the first quarter of 2012 to the fourth quarter of 2014, spanning the full treatment period prior to the 2015 regulatory changes described in section 2.

The analysis is restricted to private sector workers, defined as employees with or without a formal labour contract (*carteira assinada*). Public sector employees, military personnel, domestic workers, employers, and unpaid family workers are excluded from the sample. The self-employed are also excluded, as self-employment represents a distinct labour market category – workers operating on their own account outside the employee-employer relationship – that is not the focus of this paper. Together, private sector workers account for approximately 72% of all PNADc observations. The sample is further restricted to employed individuals – since sector of employment – the variable used to assign treatment, is not observed for unemployed workers in PNADc.

The main outcome variable is the sector-level informality share, defined as the ratio of informal private sector employees – those without a formal labour contract — to all private sector employees, weighted by individual PNADc sampling weights. Formally, for sector s and period t :

$$\text{informal_w}_{st} = \frac{\sum_{i \in s,t} \mathbb{1}(\text{informal}_i) \cdot w_i}{\sum_{i \in s,t} \mathbb{1}(\text{employee}_i) \cdot w_i}$$

where w_i is the individual sampling weight provided by PNADc. The data are then collapsed to the sector-quarter level, yielding a panel of sector-quarter observations used in all

estimations.

Treatment is assigned based on sector eligibility under the ISIC criterion, as described in Section 2. Sectors treated in the December 2011 wave are excluded from the sample, as PNADc begins in 2012 and these sectors have no observable pre-treatment periods. This exclusion is inconsequential in practice, as December 2011 cohort workers account for less than 1% of private sector workers in the sample. The final sample contains two treated cohorts: cohort 3, comprising sectors first treated in September 2012 (period 3 of the sample)², and cohort 7, comprising sectors first treated in July 2013 (period 7)³, each with 4 sectors, alongside a never-treated control group of 212 sectors. While the small number of treated sectors may appear limited, estimation draws on the full individual-level variation within each sector, corresponding to 55,848 and 286,053 individual observations for cohort 3 and cohort 7 respectively, and 1,803,637 for the never-treated group. Figure A.1 illustrates the staggered rollout of the reform across the sample period. Table 1 reports descriptive statistics by treatment group.

Table 1: Descriptive Statistics by Treatment Group

	Never-Treated	Cohort 3	Cohort 7
Number of sectors	212	4	4
Sector-quarter observations	2518	48	48
Mean informality rate	0.161	0.125	0.175
Pre-treatment informality rate	0.161	0.142	0.180
Mean population weight	264,971	469,125	2,261,459
Number of workers	1,803,637	55,848	286,053

Note: Mean informality rate is the population-weighted share of private sector employees without a formal labour contract. Pre-treatment informality uses only periods before treatment onset for each cohort. Mean population weight is the average sum of individual PNADc sampling weights per sector-quarter cell.

Two data limitations are worth noting. First, restricting the sample to employed workers is potentially endogenous, since employment itself can be an outcome of the policy. If the

²Cohort 3 comprises accommodation, water transport, air transport, and road freight transport

³Cohort 7 comprises construction of buildings, infrastructure construction, specialised construction services, and road passenger transport. The reform granted eligibility to construction firms on a site-specific rather than firm-level basis; see Section 4 for a discussion of the identification implications.

reform attracted new workers into treated sectors, the post-treatment composition of the employed sample may differ from the pre-treatment composition. Moreover, if the reform drew workers from unemployment or inactivity into formal employment, the informality share would mechanically decline as the denominator expands, independently of any genuine formalisation. Second, because PNADc is not a balanced longitudinal panel and sector of employment is unobserved for unemployed workers, worker transitions between employment and unemployment cannot be fully tracked. This means that an observed decline in informality cannot be unambiguously attributed to formalisation – it may also reflect informal workers moving into unemployment, which would not be captured in our outcome measure.

4 Methodology

This paper exploits the staggered rollout of the *desoneração da folha* across sectors to estimate the effect of payroll tax relief on labour market informality. The unit of observation is the sector-quarter cell, and the outcome of interest is the sector-level informality share constructed as described in Section 3. Treatment is assigned at the sector level based on ISIC eligibility, as described in Section 2.

Identification. The identification strategy relies on a difference-in-differences (DiD) framework that compares the evolution of informality in treated sectors relative to untreated sectors before and after treatment onset. The baseline event study specification takes the form:

$$Y_{st} = \alpha_s + \lambda_t + \sum_{k \neq -1} \beta_k \cdot \mathbb{1}(t - g_s = k) + \varepsilon_{st}$$

where Y_{st} is the informality share in sector s at time t , α_s and λ_t are sector and time fixed effects, g_s is the period in which sector s first receives treatment, and k indexes time relative to treatment onset. The coefficients β_k trace the dynamic path of treatment effects under the TWFE specification, with $k = -1$ normalised to zero as the reference period. As discussed below, the preferred estimators recover the underlying group-time average treatment

effects $ATT(g, t)$ directly, avoiding the potentially problematic weighting implicit in the β_k coefficients.

The key identifying assumption is parallel trends: in the absence of the reform, treated and untreated sectors would have followed similar trends in informality. Sectors differ structurally in labour intensity, exposure to international competition, and baseline informality rates — differences in levels that are absorbed by sector fixed effects. What the assumption requires is not that sectors are structurally identical, but that their informality trends would have evolved similarly in the absence of the reform. Since treatment eligibility was determined by a political process that did not appear to target sector-specific trends (Baumgartner et al., 2022; Lobel, 2024), this assumption is plausible. This paper tests it empirically by examining pre-treatment event study coefficients, which are all small and statistically indistinguishable from zero across all specifications, providing no evidence of differential pre-trends between treated and control sectors⁴.

The control group consists of sectors that are either never-treated or not-yet-treated at each point in time, following Callaway and Sant’Anna (2021) and Borusyak et al. (2024). The use of not-yet-treated sectors as controls increases the size of the comparison group and improves precision, under the assumption that these sectors do not anticipate their own future treatment. Additionally, PNADc does not allow observation of firm tax regime. Workers employed at firms in the simplified tax regime (*Simples Nacional*), which are not subject to payroll taxes and therefore unaffected by the reform, are included in the treated sample if they work in eligible sectors. This reinforces the intent-to-treat interpretation of the estimates and likely attenuates effects toward zero.

A further identification concern applies specifically to cohort 7, which is dominated by construction sectors. The payroll tax cut granted eligibility to construction firms based on individual construction sites rather than at the firm level, meaning treatment intensity varied

⁴Standard errors are clustered at the sector level throughout. With only 8 treated sectors, the asymptotic properties of cluster-robust standard errors may not fully apply, and inference should be interpreted with appropriate caution.

across firms and sites within eligible construction sectors. If site-level take-up was correlated with firm or site characteristics – such as size, productivity, or baseline formality rates – this introduces a potential source of endogeneity in cohort 7 estimates beyond the intent-to-treat concern that applies to all cohorts. Since cohort 7 contributes disproportionately to the weighted overall estimate, this concern extends to the aggregate results. The direction of the resulting bias depends on which construction sites took up the benefit, which cannot be determined from the available data. Cohort 7 results, and by extension the overall estimates, should therefore be interpreted with additional caution.

Staggered DiD and estimator choice. The presence of two treatment cohorts entering at different times raises well-known concerns about the use of the canonical two-way fixed effects (TWFE) estimator. [Goodman-Bacon \(2021\)](#) shows that in staggered adoption settings, TWFE produces a weighted average of all possible 2×2 DiD comparisons, including comparisons that use already-treated units as controls – so-called forbidden comparisons. When treatment effects are heterogeneous across cohorts or over time, these implicit negative weights can contaminate the overall estimate. The cohort-specific event studies presented in [Section 5](#) reveal some heterogeneity in the dynamic treatment effects across cohorts, motivating the use of modern estimators that recover the $ATT(g, t)$ ’s directly and aggregate them with well-defined, non-negative weights.

The preferred estimator follows [Borusyak et al. \(2024\)](#), henceforth BJS, who propose an imputation estimator that avoids the negative weighting problem by first estimating counterfactual outcomes under a parallel trends assumption using only untreated observations, and then imputing treated potential outcomes to recover treatment effect estimates. Sector-quarter cells are weighted by the sum of individual PNADc sampling weights, ensuring that larger sectors contribute proportionally more to the estimated treatment effect and yielding estimates representative of the average worker in treated sectors rather than the average sector.

Aggregation and reporting. Throughout, three sets of estimates are reported following the aggregation framework of [Callaway and Sant’Anna \(2021\)](#): the event study tracing dynamic effects for each period relative to treatment onset, the overall ATT summarising the average post-treatment effect, and cohort-specific ATTs assessing heterogeneity across the two treatment waves. The event study reports $\theta_{es}(e)$, defined as:

$$\theta_{es}(e) = \sum_{g \in \mathcal{G}} \mathbf{1}\{g + e \leq T\} P(G = g \mid G + e \leq T) \cdot ATT(g, g + e)$$

which averages the group-time treatment effects $ATT(g, t)$ across cohorts for each length of exposure $e = t - g$, weighted by cohort size. Unlike the β_k coefficients from the TWFE specification, this aggregation uses only non-negative weights and has a clear causal interpretation. The overall ATT reports θ_{sel}^O , which first computes the average effect for each cohort across all its post-treatment periods and then averages across cohorts weighted by group size – an aggregation whose interpretation directly parallels the ATT in the canonical two-period, two-group DiD setup. Cohort-specific ATTs $\theta_{sel}(\tilde{g})$ are also reported separately to assess treatment effect heterogeneity across the two waves. One caveat applies to the event study aggregation: as noted by [Callaway and Sant’Anna \(2021\)](#), $\theta_{es}(e)$ can be subject to compositional changes across event times when cohorts have different numbers of post-treatment periods. Since cohort 3 has more post-treatment periods than cohort 7, comparisons of $\theta_{es}(e)$ across later event times should be interpreted with this caveat in mind.

Robustness. The preferred BJS estimates are benchmarked against three alternative estimators. First, [Callaway and Sant’Anna \(2021\)](#), henceforth CS, who propose a doubly-robust estimator that computes group-time average treatment effects for each group-period pair. Second, [de Chaisemartin and D’Haultfoeuille \(2024\)](#), henceforth dCdH, who propose an estimator based on first-differencing that is robust to heterogeneous treatment effects. Third, the canonical TWFE estimator is included as a benchmark. Despite the theoretical concerns discussed above, TWFE produces broadly similar aggregate estimates in this setting, sug-

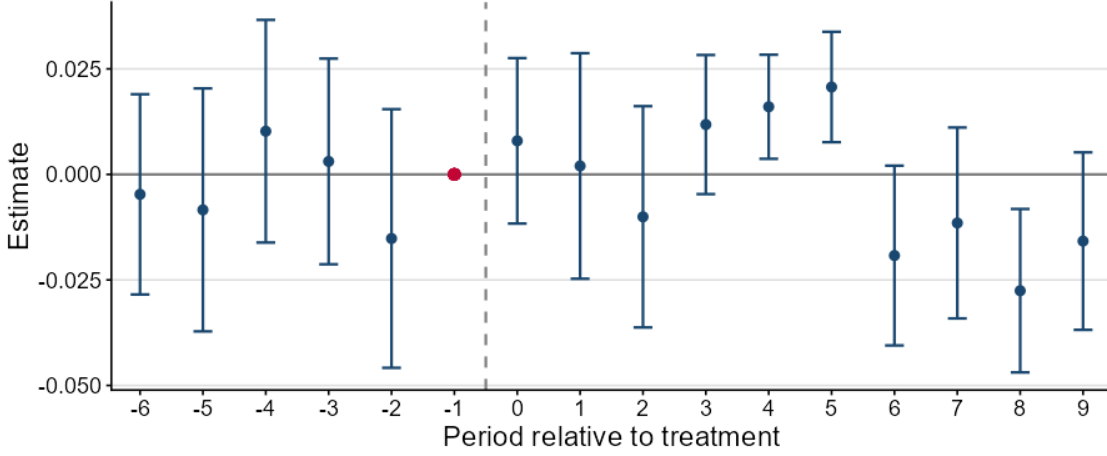
gesting that heterogeneity-induced bias does not substantially distort the overall ATT, even though cohort-specific dynamics differ. All robustness estimations use the same sample and weighting as the preferred specification. An additional robustness check compares weighted and unweighted BJS estimates to assess sensitivity to the population weighting choice.

5 Results

Main results. Figure 1 presents the overall event study estimates from the preferred BJS weighted specification. The estimates measure the change in the informality share in treated sectors relative to the control group, expressed in percentage points (pp). Pre-treatment coefficients for periods $k \leq -2$ are all small and statistically indistinguishable from zero, supporting the parallel trends assumption and providing no evidence of anticipation effects. The post-treatment pattern reveals a dynamic treatment effect that evolves substantially over time. Effects in the immediate post-treatment periods (0 to 3) are small and statistically insignificant, suggesting that firms do not immediately act on the incentive to adjust their workforce composition in response to the payroll tax cut. This is consistent with the presence of adjustment costs in labour market responses to tax changes, as formalisation decisions involve fixed entry costs and compliance burdens that firms may not incur immediately (Ulyssea, 2018; Maloney, 2004). Effects become positive and statistically significant at periods 4 and 5, peaking at approximately 1.6 and 2.1 pp respectively. From period 6 onwards, coefficients turn negative, with a statistically significant effect of approximately -2.8 pp at period 8. The overall ATT is positive but statistically insignificant (estimate: 0.5 pp), reflecting the offsetting dynamics between the positive peak and the subsequent negative reversion.

The positive effects at periods 4 and 5 indicate an increase in the informality share in treated sectors relative to the control group during these periods. While statistically significant at these specific periods, the overall ATT is not statistically distinguishable from zero,

Figure 1: Event study estimates



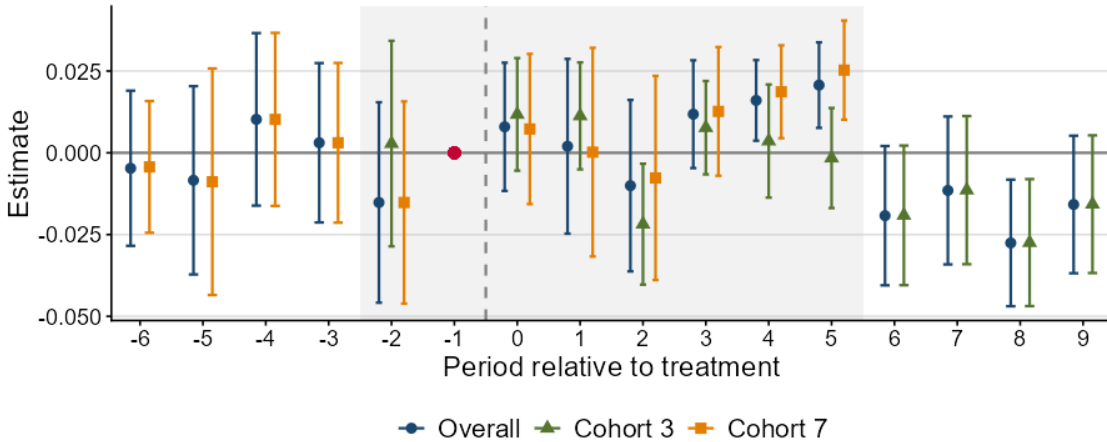
reflecting the offsetting dynamics between the positive peak and the subsequent negative reversion. Given the data limitations discussed in Section 3, and the inability to track worker transitions between employment and unemployment, these estimates should be interpreted as descriptive of the evolution of the informality share among the employed rather than as evidence of a causal shift in the formalisation margin. The direction of the effect – a temporary increase in informality share followed by a reversion – is somewhat unexpected given that the reform reduced the cost of formal employment, and disentangling the underlying mechanisms is beyond the scope of this paper. An additional channel worth noting is that positive employment effects documented in related work (Baumgartner et al., 2022; Lobel, 2024) could mechanically affect the informality share: if the reform drew workers from unemployment into formal employment, the denominator of the informality share would expand, potentially influencing the estimated coefficients independently of any compositional shift among existing workers.

The late negative reversion from period 6 onwards is intriguing but difficult to interpret with confidence. It could reflect a genuine delayed formalisation effect as firms gradually comply with the new tax regime, broader macroeconomic deterioration in Brazil during 2013-2014 that affected labour market composition differentially across sectors, or other sector-specific dynamics. Whether this pattern represents a true reversal of the earlier effect

or reflects factors unrelated to the reform remains an open question. Notably, as discussed further in the heterogeneity analysis below, this late reversion is entirely attributable to cohort 3.

Treatment effect heterogeneity. Figure 2 presents cohort-specific event study estimates alongside the overall series. For cohort 7, pre-treatment coefficients are small and insignificant across the five available pre-treatment periods, supporting the parallel trends assumption. For cohort 3, only one pre-treatment coefficient can be estimated (at $k = -2$, since $k = -1$ is the reference period), which limits the empirical basis for assessing parallel trends for this cohort, a limitation worth bearing in mind when interpreting its post-treatment estimates.

Figure 2: Event study estimates (overall and cohort-specific)



The two cohorts show positive effects in early post-treatment periods but diverge notably in later periods, revealing meaningful treatment effect heterogeneity. Cohort 7 displays an initial negative dip at period 2, followed by a positive and increasing pattern from period 3 onwards, peaking at periods 4 and 5 with estimates of approximately 1.9 and 2.5 pp, the only statistically significant coefficients for this cohort. Cohort 3 exhibits a more complex pattern: a negative dip at period 2, followed by a recovery at periods 3 and 4, before turning persistently negative from period 6 onwards. Most cohort 3 coefficients are individually

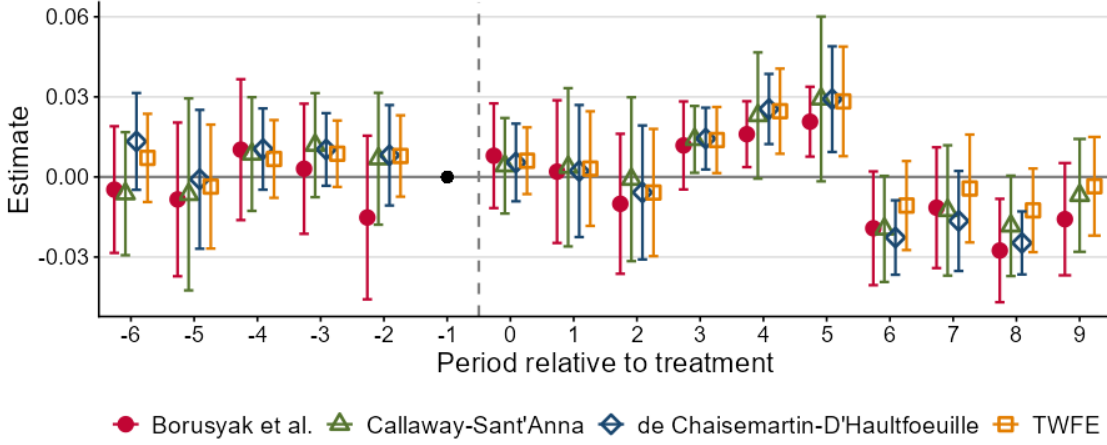
insignificant, with the exception of a significant negative dip at period 2 and a significant negative coefficient at period 8, which tempers any strong interpretation of the overall dynamic pattern.

The late negative reversion in the overall series from period 6 onwards is entirely attributable to cohort 3. Cohort 7 enters treatment at period 7 and the sample ends at period 12, leaving only five post-treatment periods in relative time (periods 0 to 5) – the overall series beyond period 5 therefore reflects cohort 3 exclusively. The overall event study series (estimated on the full sample across both cohorts) tracks cohort 7 more closely in early post-treatment periods, consistent with the substantially larger population weight of cohort 7 relative to cohort 3. As shown in Table 1, the mean population weight of cohort 7 sectors is approximately five times that of cohort 3 sectors, meaning cohort 7 contributes disproportionately to the weighted overall estimate.

Robustness. Figure 3 presents the event study estimates from all four estimators – BJS, CS, dCdH, and TWFE – on the same axes. The four estimators produce broadly consistent results in terms of direction and timing. Pre-treatment coefficients are small and insignificant across all specifications, further supporting the parallel trends assumption. The positive effects at periods 3 to 5 and the subsequent negative reversion are visible across all estimators, with the main differences being in the width of confidence intervals and the precise periods at which individual coefficients cross the significance threshold.

The pattern of period-specific significance varies across estimators, and the inconsistency in which periods are significant across specifications suggests that individual event study coefficients should not be interpreted as robust causal evidence of an effect on informality. The overall ATT estimates, reported in Table 2, are positive across all five specifications but statistically insignificant. This consistency in direction but lack of statistical significance is the central finding of this paper, and is consistent with the finding in Lobel (2024) that informality is not a major margin of response to the Brazilian payroll tax reform. Cohort-

Figure 3: Event study estimates across alternative estimators



specific ATT estimates further reveal that this overall null reflects opposing dynamics rather than a uniform zero effect across both cohorts: cohort 7 ATTs are positive and significant under some specifications while cohort 3 ATTs are negative. The dynamic pattern visible in the event study plots warrants further investigation, but its causal interpretation remains uncertain given the limitations discussed throughout the paper.

Table 2: ATT by Treatment Group

	BJS	BJS (unw.)	CS	dCdH	TWFE
Overall	0.005 (0.008)	0.002 (0.007)	0.009 (0.007)	0.009 (0.005)	0.006 (0.009)
Cohort 3	-0.007 (0.010)	-0.014 (0.008)	-0.007 (0.005)	-0.010** (0.004)	-0.013 (0.007)
Cohort 7	0.009 (0.009)	0.007 (0.009)	0.016** (0.006)	0.015** (0.007)	0.008 (0.008)

Note: Point estimates with standard errors in parentheses, clustered at the sector level. Overall ATT is the weighted average treatment effect across all treated cohorts. Cohort 3 and Cohort 7 refer to sectors first treated in periods 3 and 7 respectively. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The close alignment between TWFE and modern estimator estimates is also consistent with the theoretical observation that the negative weighting problem is less severe when treated units represent a small fraction of the sample (de Chaisemartin and D'Haultfoeulle,

2020; Goodman-Bacon, 2021). With only 8 treated sectors against 212 never-treated ones, the control group in this paper is overwhelmingly composed of clean comparisons, limiting the scope for contamination from already-treated controls.

Figure A.4 compares weighted and unweighted BJS estimates. The two series are broadly consistent in direction and significance, with some minor divergence in early post-treatment periods that does not affect the main conclusions. This suggests that the main findings are not driven by the population weighting choice.

6 Conclusion

This paper estimates the effect of the *desoneração da folha*, Brazil’s 2011 payroll tax relief program, on labour market informality at the sector level, using microdata from the PNADc household survey and a staggered difference-in-differences design. Exploiting the staggered rollout of the reform across sectors as a source of identifying variation, the preferred BJS imputation estimator is applied to a panel of sector-quarter observations covering twelve quarters from 2012 to 2014, with results benchmarked against three alternative estimators that differ in how they aggregate treatment effects and handle heterogeneity across cohorts.

The central finding is that the reform did not produce a statistically significant average effect on the informality share. Overall ATT estimates are positive but insignificant across all five specifications considered, and this consistency across estimators strengthens confidence in the null result. This finding is consistent with the broader literature suggesting that informality is not a primary margin of adjustment to payroll tax changes in Brazil (Baumgartner et al., 2022; Lobel, 2024). While some individual event study coefficients are statistically significant at specific periods, the inconsistency in which periods are significant across estimators, combined with the insignificant overall ATT, cautions against a causal interpretation of these period-specific findings.

The cohort-specific analysis reveals meaningful treatment effect heterogeneity. Cohort-

specific ATT estimates, reported in Table 2, confirm that the two treated cohorts point in opposite directions – cohort 7 ATTs are positive, while cohort 3 ATTs are negative – and the overall null result reflects their weighted average rather than a genuine zero effect for both cohorts. The late negative reversion in the full sample event study is entirely attributable to cohort 3, whose longer post-treatment window extends beyond the five periods available for cohort 7. Whether these cohort-specific patterns reflect genuine heterogeneity in the reform’s effects across sectors, or are driven by broader macroeconomic dynamics in 2013–2014, remains an open question.

Several limitations should be noted. First, the analysis is restricted to employed workers, since sector of employment is not observed for unemployed individuals in PNADc. This restriction is potentially endogenous if the reform affected employment levels, and means that transitions between employment and unemployment cannot be tracked: a decline in informality share cannot be unambiguously attributed to formalisation, as it may also reflect mechanical changes in the composition of the employed sample. Second, the small number of treated sectors (four per cohort) may limit the precision of sector-clustered standard errors, as cluster-robust inference relies on asymptotic arguments that perform best with a larger number of clusters.

This paper contributes to the literature on the labour market effects of payroll taxation in developing economies by providing sector-level evidence on informal employment composition using household survey data, complementing firm-level analyses that rely on formal sector administrative records. The null result on the aggregate informality share is consistent with existing evidence, and the dynamic and cohort-specific patterns identified here suggest that the relationship between payroll costs and informal employment warrants further investigation with longer time horizons and more granular data. A natural extension would be to examine the reform’s effect on the wage differential between formal and informal workers within treated sectors, which would speak in more detail to the incidence of the tax cut along the informality margin and could be implemented within the same staggered difference-in-

differences framework. A key challenge for such an analysis is that the minimum wage in Brazil tends to bind for unskilled workers in both the formal and informal sectors ([Meghir et al., 2015](#); [Haanwinckel and Soares, 2021](#)), compressing the formal-informal wage differential across all sectors and potentially limiting the variation available to detect reform-induced changes in the wage gap.

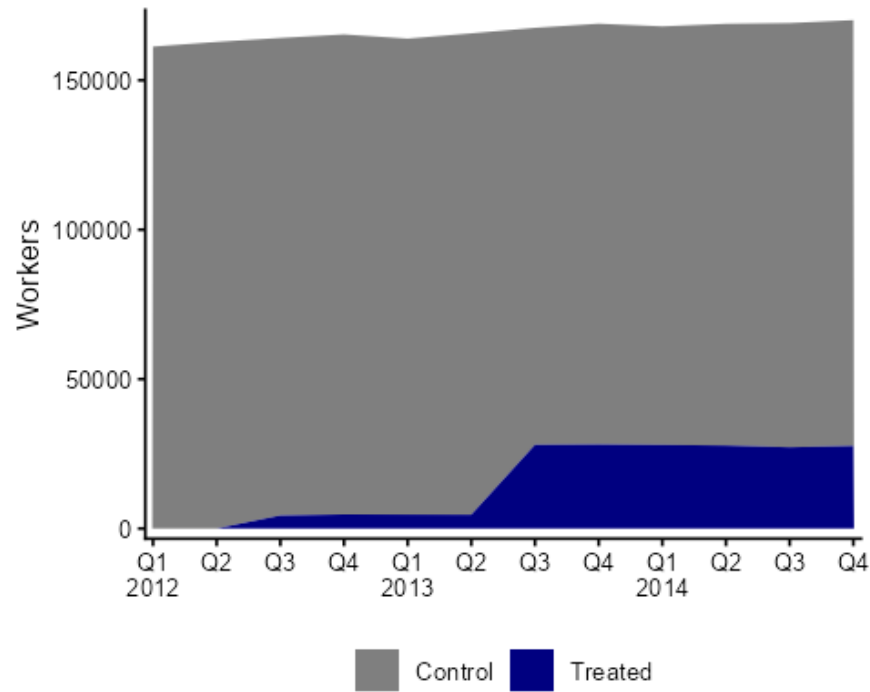
References

- BAUMGARTNER, E., R. CORBI, AND R. NARITA (2022): “Payroll Tax, Employment and Labor Market Concentration,” Working paper.
- BORUSYAK, K., X. JARAVEL, AND J. SPIESS (2024): “Revisiting Event-Study Designs: Robust and Efficient Estimation,” *The Review of Economic Studies*, 91, 3253–3285.
- CALLAWAY, B. AND P. H. SANT’ANNA (2021): “Difference-in-Differences with multiple time periods,” *Journal of Econometrics*, 225, 200–230.
- DE CHAISEMARTIN, C. AND X. D’HAULTFŒUILLE (2020): “Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects,” *American Economic Review*, 110, 2964–2996.
- (2024): “Difference-in-Differences Estimators of Intertemporal Treatment Effects,” *The Review of Economics and Statistics*, 1–45.
- GOODMAN-BACON, A. (2021): “Difference-in-differences with variation in treatment timing,” *Journal of Econometrics*, 225, 254–277.
- GRUBER, J. (1997): “The Incidence of Payroll Taxation: Evidence from Chile,” *Journal of Labor Economics*, 15, S72–S101.
- HAANWINCKEL, D. AND R. R. SOARES (2021): “Workforce Composition, Productivity, and Labour Regulations in a Compensating Differentials Theory of Informality,” *The Review of Economic Studies*, 88, 2970–3010.
- KUGLER, A. AND M. KUGLER (2009): “Labor Market Effects of Payroll Taxes in Developing Countries: Evidence from Colombia,” *Economic Development and Cultural Change*, 57, 335–358.

- KUGLER, A., M. KUGLER, AND L. O. HERRERA PRADA (2017): “Do Payroll Tax Breaks Stimulate Formality? Evidence from Colombia’s Reform,” Working Paper 23308, National Bureau of Economic Research.
- LOBEL, F. (2024): “Who Benefits from Payroll Tax Cuts? Market Power, Tax Incidence and Efficiency,” Revised and resubmitted, *Journal of Political Economy*.
- MALONEY, W. F. (2004): “Informality Revisited,” *World Development*, 32, 1159–1178.
- MEGHIR, C., R. NARITA, AND J.-M. ROBIN (2015): “Wages and Informality in Developing Countries,” *American Economic Review*, 105, 1509–1546.
- MEYER, L., C. LUCINDA, AND H. SPOLADOR (2021): “The payroll tax exemption in Brazil: structural and quasi-experimental perspectives,” *Revista Brasileira de Economia*, 75, 37–63.
- SAEZ, E., B. SCHOEFER, AND D. SEIM (2019): “Payroll Taxes, Firm Behavior, and Rent Sharing: Evidence from a Young Workers’ Tax Cut in Sweden,” *American Economic Review*, 109, 1717–1763.
- ULYSSEA, G. (2018): “Firms, Informality, and Development: Theory and Evidence from Brazil,” *American Economic Review*, 108, 2015–2047.
- (2020): “Informality: Causes and Consequences for Development,” *Annual Review of Economics*, 12, 525–546.

A Figures and Tables

Figure A.1: Staggered rollout of the payroll tax cut



Note: The figure shows the number of workers in treated and never-treated sectors by quarter, weighted by individual PNADc sampling weights. Treated workers are those employed in sectors eligible under the ISIC criterion.

Figure A.2: ATT across estimators

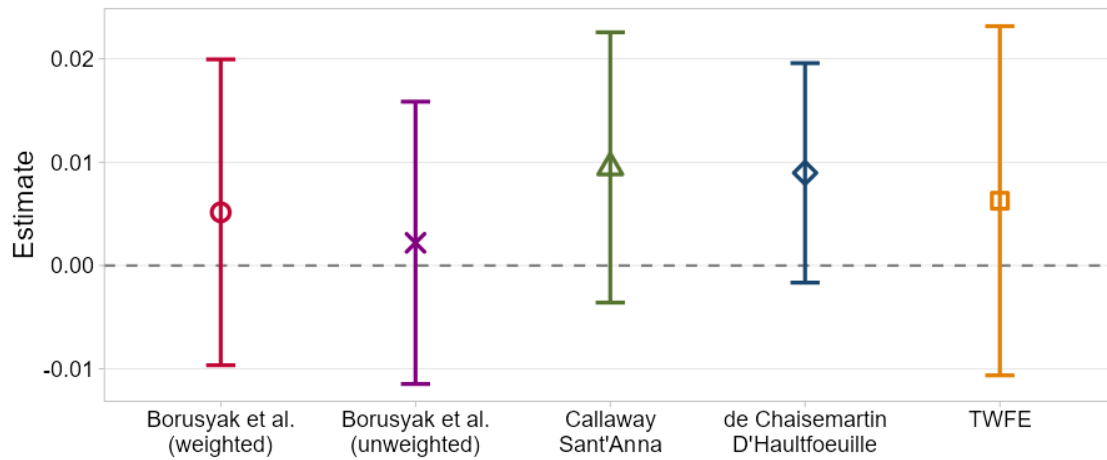


Figure A.3: Event study estimates by cohort, calendar time

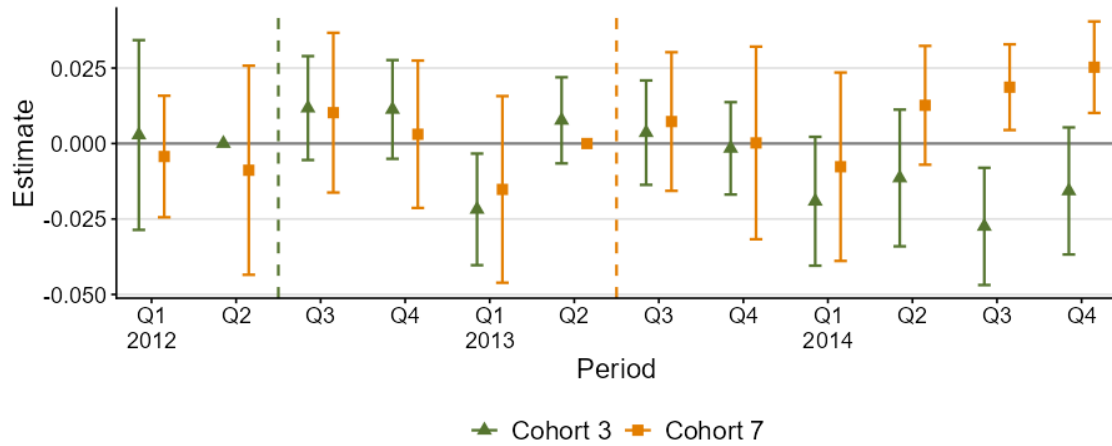


Figure A.4: Event study estimates (weighted vs. unweighted)

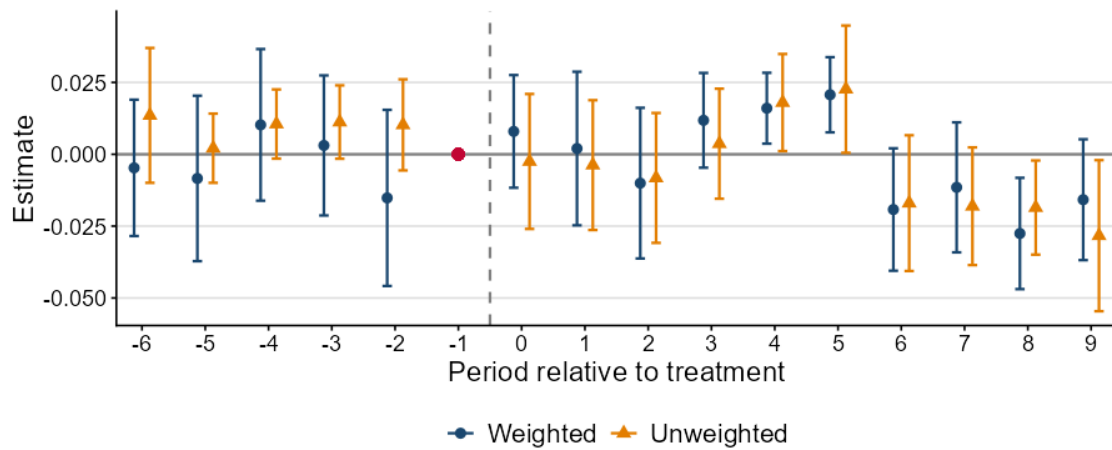


Table A.1: Event study estimates (overall sample)

Time to treatment	BJS	BJS (unw.)	CS	dCdH	TWFE
−6 quarters	−0.005 (0.012)	0.014 (0.012)	−0.006 (0.009)	0.013 (0.009)	0.007 (0.008)
−5 quarters	−0.008 (0.015)	0.002 (0.006)	−0.007 (0.015)	−0.001 (0.013)	−0.004 (0.012)
−4 quarters	0.010 (0.013)	0.011* (0.006)	0.009 (0.009)	0.010 (0.008)	0.007 (0.007)
−3 quarters	0.003 (0.012)	0.011* (0.007)	0.012 (0.008)	0.010 (0.007)	0.009 (0.006)
−2 quarters	−0.015 (0.016)	0.010 (0.008)	0.007 (0.010)	0.008 (0.010)	0.008 (0.008)
0 (treatment)	0.008 (0.010)	−0.003 (0.012)	0.004 (0.007)	0.005 (0.007)	0.006 (0.006)
+1 quarter	0.002 (0.014)	−0.004 (0.012)	0.004 (0.012)	0.002 (0.013)	0.003 (0.011)
+2 quarters	−0.010 (0.013)	−0.008 (0.012)	−0.001 (0.013)	−0.006 (0.013)	−0.006 (0.012)
+3 quarters	0.012 (0.008)	0.004 (0.010)	0.014*** (0.005)	0.014** (0.006)	0.014** (0.006)
+4 quarters	0.016** (0.006)	0.018** (0.009)	0.023** (0.010)	0.025*** (0.007)	0.025*** (0.008)
+5 quarters	0.021*** (0.007)	0.023** (0.011)	0.029** (0.013)	0.029*** (0.010)	0.028*** (0.010)
+6 quarters	−0.019* (0.011)	−0.017 (0.012)	−0.019** (0.008)	−0.023*** (0.007)	−0.011 (0.009)
+7 quarters	−0.012 (0.012)	−0.018* (0.010)	−0.013 (0.010)	−0.016* (0.010)	−0.004 (0.010)
+8 quarters	−0.028*** (0.010)	−0.019** (0.008)	−0.018** (0.008)	−0.025*** (0.006)	−0.013 (0.008)
+9 quarters	−0.016 (0.011)	−0.028** (0.013)	−0.007 (0.009)		−0.003 (0.009)

Note: Point estimates with standard errors in parentheses, clustered at the sector level. Period relative to treatment is measured in quarters. Period 0 denotes the first period of treatment and period −1 is the omitted reference period. dCdH does not produce an estimate for period 9. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.2: Event study estimates (cohort 3)

Time to treatment	BJS	BJS (unw.)	CS	dCdH	TWFE
−2 quarters	0.003 (0.016)	−0.003 (0.005)	0.002 (0.010)	0.006 (0.010)	0.005 (0.009)
0 (treatment)	0.012 (0.009)	−0.000 (0.008)	0.013** (0.006)	0.011* (0.006)	0.011** (0.005)
+1 quarter	0.011 (0.008)	0.000 (0.009)	0.004 (0.005)	0.001 (0.004)	0.001 (0.004)
+2 quarters	−0.022** (0.009)	−0.013* (0.007)	−0.017** (0.007)	−0.020*** (0.007)	−0.020*** (0.006)
+3 quarters	0.008 (0.007)	−0.013** (0.006)	0.002 (0.007)	−0.001 (0.008)	−0.001 (0.007)
+4 quarters	0.004 (0.009)	−0.011 (0.007)	−0.009 (0.008)	−0.008 (0.007)	−0.008 (0.006)
+5 quarters	−0.002 (0.008)	−0.016*** (0.005)	−0.008 (0.007)	−0.007 (0.005)	−0.008 (0.005)
+6 quarters	−0.019* (0.011)	−0.017 (0.012)	−0.019** (0.008)	−0.023*** (0.007)	−0.023*** (0.007)
+7 quarters	−0.011 (0.012)	−0.018* (0.010)	−0.013 (0.010)	−0.016* (0.010)	−0.017* (0.009)
+8 quarters	−0.027*** (0.010)	−0.019** (0.008)	−0.018** (0.007)	−0.025*** (0.006)	−0.024*** (0.006)
+9 quarters	−0.016 (0.011)	−0.028** (0.013)	−0.007 (0.008)		−0.015** (0.008)

Note: Point estimates with standard errors in parentheses, clustered at the sector level. Period relative to treatment is measured in quarters. Period 0 denotes the first period of treatment and period −1 is the omitted reference period. dCdH does not produce an estimate for period 9. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.3: Event study estimates (cohort 7)

Time to treatment	BJS	BJS (unw.)	CS	dCdH	TWFE
−6 quarters	−0.004 (0.010)	0.015 (0.011)	−0.006 (0.009)	0.013 (0.009)	0.010 (0.009)
−5 quarters	−0.009 (0.018)	0.004 (0.007)	−0.007 (0.015)	−0.001 (0.013)	−0.000 (0.012)
−4 quarters	0.010 (0.014)	0.012* (0.006)	0.009 (0.009)	0.010 (0.008)	0.010 (0.007)
−3 quarters	0.003 (0.012)	0.013* (0.007)	0.012 (0.008)	0.010 (0.007)	0.012* (0.006)
−2 quarters	−0.015 (0.016)	0.013 (0.010)	0.008 (0.012)	0.009 (0.011)	0.009 (0.010)
0 (treatment)	0.007 (0.012)	−0.003 (0.014)	0.003 (0.009)	0.005 (0.009)	0.005 (0.008)
+1 quarter	0.000 (0.016)	−0.004 (0.014)	0.004 (0.014)	0.003 (0.015)	0.003 (0.014)
+2 quarters	−0.008 (0.016)	−0.007 (0.014)	0.003 (0.014)	−0.003 (0.015)	−0.003 (0.014)
+3 quarters	0.013 (0.010)	0.007 (0.012)	0.017** (0.007)	0.018** (0.007)	0.017** (0.007)
+4 quarters	0.019*** (0.007)	0.024** (0.010)	0.030*** (0.010)	0.032*** (0.008)	0.032*** (0.007)
+5 quarters	0.025*** (0.008)	0.030** (0.013)	0.037*** (0.014)	0.037*** (0.012)	0.037*** (0.011)

Note: Point estimates with standard errors in parentheses, clustered at the sector level. Period relative to treatment is measured in quarters. Period 0 denotes the first period of treatment and period −1 is the omitted reference period. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.